Sustainable Human Societies

Economic growth and development

• **Economic growth** is an increase in a nation's capacity to provide goods and services to people.
It involves making an economy bigger. Its goal is to use political and economic systems to *encourage environmentally beneficial and more sustainable forms of economic development, and to discourage environmentally harmful and unsustainable forms of economic growth.*
• **Economic development** is the improvement of human living standards made possible by economic growth.

Economic Systems Are Supported by Three Types of Resources

●**An economic system is a social institution** through which goods and services are produced, distributed, and consumed to satisfy people's needs and wants, ideally in the most efficient way possible.
●Three types of capital, or resources, are used to produce goods and services.
♦**Natural capital** includes resources and services produced by the earth's natural processes, which support all economies and all life.
♦**Human capital, or human resources**, includes people's physical and mental talents that provide labor, organizational and management skills, and innovation.
♦**Manufactured capital, or manufactured resources**, refers to items such as machinery, equipment, and factories made from natural resources with the help of human resources.

Natural capital:

Environmental and ecological economists have developed various tools for estimating the values of the earth's natural capital.

The three goals of this study are to

(1) integrate economic and ecological knowledge in order to estimate the economic and ecological values of ecosystem services;

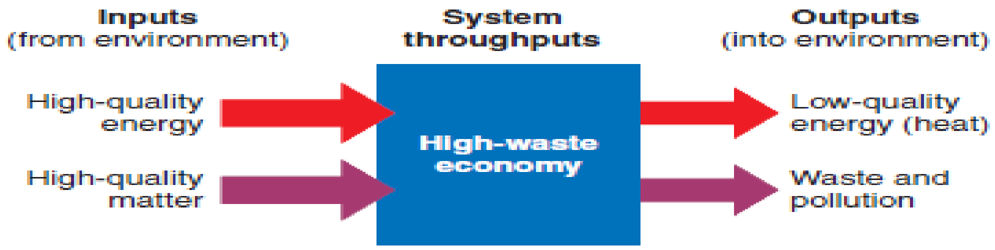
(2) to evaluate the costs and benefits of actions that could be taken to prevent the decline of these services;

(3) to develop toolkits to help local, regional, and international policy makers promote more sustainable development that conserves ecosystems and biodiversity.

Ecological and environmental economists have developed ways to estimate ***non-use values*** *of natural resources and ecological services that are not represented in market transactions. One such value is an* ***existence value****—a monetary value placed on a resource such as an old-growth forest or endangered species just because it exists, even though we may never see it or use it. Another is* ***aesthetic value****—a monetary value placed on a forest, species, or a part of nature because of its beauty. A third type, called a* ***bequest or option value****, is based on the willingness of people to pay to protect some forms of natural capital for use by future generations.*

High-throughput economies of most developed countries

• The high-throughput economies of most developed countries rely on continually increasing the rates of energy and matter flow to increase economic growth.



• This practice produces valuable goods and services, but it also converts high-quality matter and energy resources into waste, pollution, and low-quality heat, and in the process, can deplete or degrade various forms of natural capital that support all life and economies.

Environmental economists believe that conventional economic growth eventually will become unsustainable. Their reasoning is that such growth will lead us to deplete or degrade much of the natural capital.
• Ecological economists see all economies as human subsystems of the biosphere that depend on natural resources and services provided by the sun and earth. They urge us to shift from our current high throughput economies to more *economically sustainable economies, or eco-economies.*

Environmental Economic Indicators

• Economic growth is usually measured by the percentage of change in a country's **gross domestic product (GDP): the annual market** value of all goods and services produced by all firms and organizations, foreign and domestic, operating within a country
• Changes in a country's economic growth per person are measured by **per capita GDP: the GDP divided by the country's total** population at midyear.
• Environmental and ecological economists and environmental scientists call for the development and widespread use of new indicators—called *green indicators—to help monitor environmental* quality and human wellbeing.
• One such indicator is the **genuine progress indicator (GPI)**

Subsidies

• Subsidies activities and projects which prevent pollution or don't degrade environment.
• Tax the activities and projects which pollute and degrade environment.
• Japan, France, and Belgium have phased out all coal subsidies and Germany plans to do so by 2018. China has cut coal subsidies by about 73% and has imposed a tax on high-sulfur coals.
• Governments could phase in environmentally beneficial subsidies and tax breaks for pollution prevention, eco-cities, sustainable forestry, sustainable agriculture, sustainable water use, energy efficiency and renewable energy use, and actions to slow projected climate change.
• The economy also will grow because of this.

Use Lessons from Nature to Shift to More Sustainable Economies-*low-throughput economy*

Learning and applying lessons from nature can help us design and manage more sustainable economies.

A*low-throughput economy*, based on energy flow and matter recycling, works with nature to reduce excessive throughput and the unnecessary waste of matter and energy resources (green boxes).

This is done by

(1) reusing and recycling most non-renewable matter resources;

(2) using renewable resources no faster than they are replenished;

(3) reducing resource waste by using matter and energy resources more efficiently;

(4) reducing unnecessary and environmentally harmful forms of consumption;

(5) emphasizing pollution prevention and waste reduction; and

6) controlling population growth to stabilize the number of matter and energy consumers.

We can use certain principles for shifting to more environmentally sustainable economies, or *eco-economies*, during this century.

Economics:

Use full cost pricing

Sell more services instead of more things

Reduce poverty

Use eco-labels

Use environmental indicators to measure progress

Ecology and population

Mimic nature

Preserve biodiversity

Repair ecological damage

Stabilize human population.

Resource use and control pollution

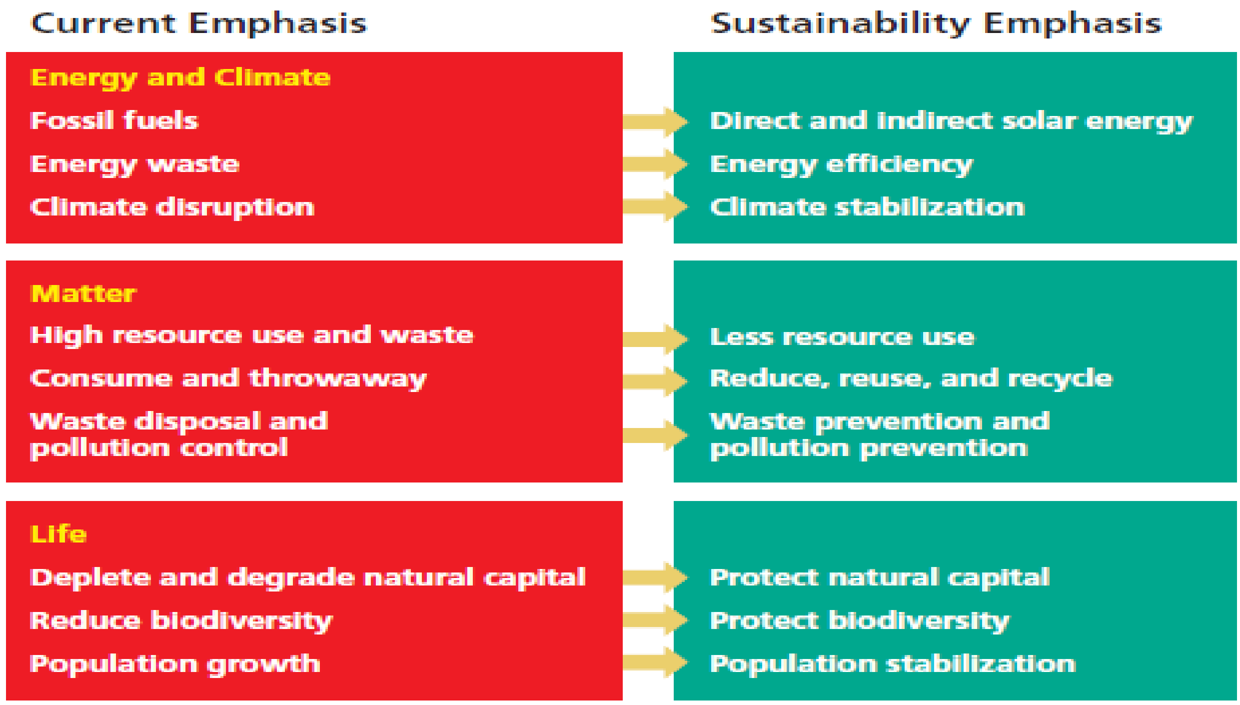
Improve energy efficiency

Rely more on renewable resources

Shift to a non-carbon renewable energy economy

The environmental revolution

These are some of the cultural shifts in emphasis that will be necessary to bring about the environmental or sustainability revolution.



Education on the ecological index is one of the key parameters in leading towards a sustainable developed nation. So let's all become socially and ecologically responsible.